

Inducing Implicit Focus through Structured Irrelevance for Robust Neural PDE Solvers

Yunpeng Gong[†], Qingyuan Zeng^{*}, Chenchu Liu[†], Jian Guan^{*}, Jiajing Lin[†], Chuangliang Zhang[†], Yongjie Hou[¶],
Yipeng Huang[†], Zhenzhong Wang^{*}, Xiaobo Li[‡], Xiaodan Feng[§], Min Jiang[†]✉

[†]School of Informatics, Xiamen University, Xiamen, China

^{*}Institute of Artificial Intelligence, Xiamen University, Xiamen, China

[¶]School of Electronic Science and Engineering, Xiamen University, Xiamen, China

[‡]Xi'an Thermal Power Research Institute Co., Ltd. 710054, China

[§]Huaneng Renewables Co., Ltd. 100036, China

Abstract—Neural solvers for partial differential equations (PDEs) have shown remarkable potential across scientific computing and physics-informed learning. However, their performance often hinges on fragile feature dependencies, leading to poor generalization and vulnerability to input noise or domain shifts. In this work, we propose a simple yet effective training-time perturbation scheme that improves generalization by injecting structured, semantically irrelevant signals into the input space. Specifically, we randomly replace small, localized regions of the input with constant values, thereby compelling the model to focus on predictive patterns that remain stable under partial information loss. We formally analyze the learning dynamics and show that such perturbations induce an implicit feature selection mechanism, which we term *Perturbation-Induced Implicit Focus (PIIF)*. This mechanism arises as the model progressively suppresses unstable or co-dependent input dimensions and prioritizes those that consistently yield informative gradients across masked configurations. Applied to neural PDE solvers, our method improves both cross-domain generalization and robustness to partial corruption, without requiring architectural changes or auxiliary supervision. The results suggest that structured irrelevance serves as an effective inductive bias, guiding neural solvers to discover physically meaningful and resilient representations.

Index Terms—Neural PDE Solvers, Representation Robustness, Implicit Attention, Scientific Machine Learning.

I. INTRODUCTION

Neural network-based solvers for partial differential equations (PDEs) have emerged as powerful tools across a broad spectrum of scientific and engineering domains, including fluid dynamics, electromagnetic simulation, and climate modeling. These models offer strong representational capacity and computational scalability, making them promising alternatives to traditional numerical schemes. However, despite their impressive empirical performance, neural PDE solvers often suffer from poor generalization when applied to inputs outside the

training distribution, such as changes in boundary conditions, resolution scales, or domain geometries.

One of the key reasons behind this fragility lies in the model's tendency to overfit to coincidental correlations in the training data rather than capturing structurally stable associations. Neural models are known to exploit surface-level patterns that may be predictive on average, yet fail to generalize when the input becomes noisy, sparse, or partially corrupted. While recent research has introduced various architectural inductive biases—such as symmetry priors, physics-informed losses, and equivariant layers—to mitigate this issue, the problem of learning *robust feature-level representations* remains underexplored.

In this work, we address this gap by proposing a simple yet principled training mechanism that injects *structured irrelevance* into the input during training. Concretely, we randomly replace a small subset of input dimensions with a fixed constant to simulate controlled information loss. These perturbations are semantically meaningless and task-agnostic, serving as structured distractions that compel the model to identify feature dimensions that maintain predictive utility even when the input is incomplete. As illustrated in Figure 1, such perturbations manifest as small, localized regions within the input domain being replaced by a constant value. This visual degradation simulates information loss and forces the model to learn predictive features that remain effective under partial occlusion. This training-time degradation process is not aimed at increasing data diversity but at forcing the model to focus on stable input-output associations that are resilient under masking.

We further provide a theoretical analysis of the resulting learning dynamics. By examining the gradient responses over time, we show that the model naturally learns to suppress input dimensions whose gradient contributions are inconsistent under perturbation, while reinforcing those dimensions that repeatedly provide reliable signals. This effect gives rise to an implicit feature filtering mechanism, which we term *Perturbation-Induced Implicit Focus (PIIF)*. Unlike explicit attention modules, PIIF emerges organically from backpropa-

Min Jiang and Yunpeng Gong are with the Department of Artificial Intelligence, Key Laboratory of Multimedia Trusted Perception and Efficient Computing, Ministry of Education of China, School of Informatics, Key Laboratory of Digital Protection and Intelligent Processing of Intangible Cultural Heritage of Fujian and Taiwan, Ministry of Culture and Tourism, Xiamen University, Xiamen 361005, Fujian, P.R. China (e-mail: minjiang@xmu.edu.cn; gongyunpeng@stu.xmu.edu.cn or fmonkey625@gmail.com).

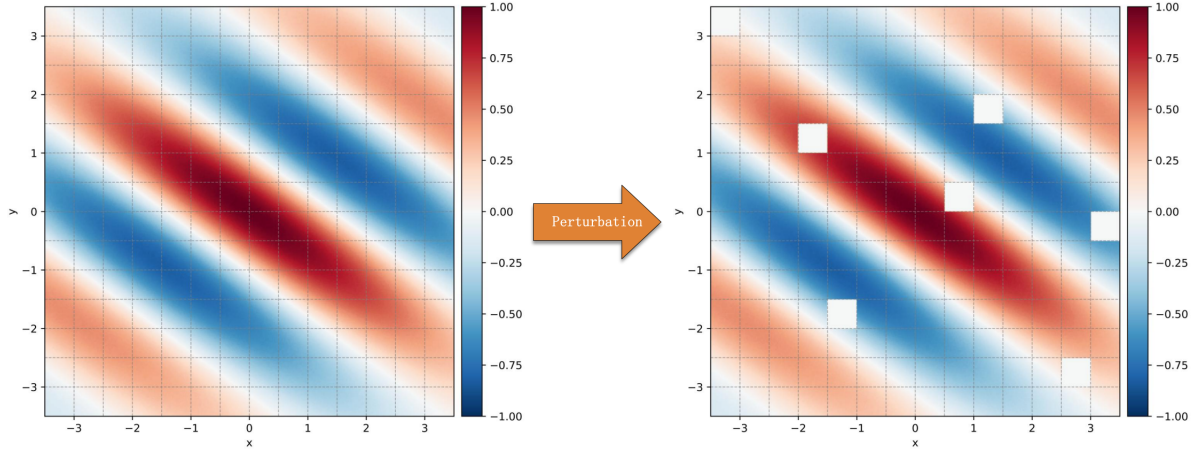


Fig. 1. The left panel illustrates a clean input field with smoothly varying features, while the right panel shows the same input after applying structured perturbations—localized regions are replaced with constant values. This perturbation simulates partial information loss, encouraging the model to learn robust and predictive patterns that persist even under incomplete or noisy conditions.

gation under stochastic masking and offers a lightweight and architecture-agnostic way to achieve implicit attention and robust generalization.

We demonstrate the effectiveness of our method on several benchmark PDE-solving tasks, showing consistent gains in generalization across domain shifts, partial observations, and corrupted boundaries. Our findings suggest that injecting structured irrelevance into the training process acts as an effective inductive prior, enabling neural PDE solvers to discover sparse, stable, and physically meaningful representations.

a) Contributions.: Our key contributions are summarized as follows:

- We introduce a structured input perturbation mechanism that promotes robustness by simulating information degradation through semantically irrelevant masking.
- We theoretically characterize the emergence of perturbation-induced implicit focus (PIIF), revealing a gradient-driven mechanism for implicit feature selection.
- We validate our approach on neural PDE solvers and demonstrate significant improvements in generalization and robustness without modifying the architecture or adding supervision.

II. RELATED WORK

a) Neural PDE Solvers: Learning-based PDE solvers have received growing attention for their ability to approximate complex systems with neural networks. Physics-Informed Neural Networks (PINNs) encode governing equations into the loss function and have shown promise for forward and inverse problems [1]. Operator learning approaches such as the Fourier Neural Operator (FNO) generalize across resolutions and geometries by learning mappings between infinite-dimensional function spaces [2]. Graph-based PDE solvers leverage message passing and spatial priors for mesh-based domains [3]. While these models improve predictive performance, they

often lack robustness to sparse or corrupted inputs and do not explicitly model feature reliability under perturbations.

b) Input Perturbation and Feature Masking: Input-level perturbations have proven effective for regularization in vision and NLP. Random Color Erasing [12], [15], [19], [21] randomly erases local color information in RGB images to prevent the model from overfitting to color features. Gong et al. [18] proposed a local feature masking strategy as a more robust alternative to dropout [4], improving CNN generalization by selectively erasing spatial features during training. In language models, masked language modeling [5] encourages contextual inference by hiding words. While these methods improve generalization, they are typically tied to auxiliary tasks or reconstruction objectives. Our method introduces structured, semantically irrelevant perturbations not to reconstruct masked content, but to shape the model’s internal attention distribution by exposing it to information loss.

c) Implicit Attention and Feature Selection: Neural attention mechanisms have enabled models to learn soft weighting over inputs [6]. Beyond explicit attention, recent research has uncovered emergent sparsity and feature focus from training dynamics [7]. Post-hoc feature attribution methods such as DeepLIFT [8] aim to interpret model decisions, but do not influence learning behavior. Our method provides a training-time mechanism that implicitly induces focus on stable input dimensions through perturbation-driven gradient filtering, without requiring additional modules or losses.

d) Robustness and Inductive Bias in Scientific Learning: Improving generalization under distributional shift is critical for scientific ML applications. Some approaches enforce physical constraints via architecture or symbolic priors [9], while others use noise injection to simulate measurement corruption [10]. Inductive biases such as symmetry, equivariance, and compositionality are increasingly emphasized in model design [11], [17]. Our work complements these efforts by introducing a lightweight input-level bias that encourages the model to

prioritize structurally reliable features through a theoretically grounded perturbation process.

III. METHOD

We propose a training-time perturbation strategy for neural PDE solvers that injects structured irrelevance into the input space to induce robust generalization and implicit feature selection. Unlike conventional data augmentation, our method does not aim to increase data diversity, but rather introduces controlled information degradation to simulate scenarios where parts of the input may be unreliable, noisy, or missing. This encourages the model to identify and rely on features that remain predictive even under incomplete observations.

Let $x \in \mathbb{R}^d$ denote the input vector, which may include spatial-temporal coordinates, boundary values, or physical measurements depending on the PDE task. We define a stochastic masking operator $\mathcal{M}_p$ that replaces each input dimension x_i independently with a fixed constant $c \in \mathbb{R}$ (such as zero or the empirical mean) with probability $p \in (0, 1)$. Formally, the perturbed input $\tilde{x}$ is constructed as:

$$\tilde{x}_i = \begin{cases} x_i, & \text{with probability } 1 - p \\ c, & \text{with probability } p \end{cases}, \quad i = 1, 2, \dots, d.$$

This procedure is applied only during training. The perturbation is semantically meaningless yet structured, operating at the dimension level and introducing a form of localized, controlled irrelevance.

To understand the mechanism by which this perturbation improves robustness, we analyze the gradient dynamics induced by this training regime. Let f_θ be the neural solver and $\mathcal{L}(f_\theta(\tilde{x}), y)$ the supervised loss on perturbed input. We denote the instantaneous gradient magnitude with respect to input dimension i at iteration t as $g_i^{(t)} := \left\| \frac{\partial \mathcal{L}^{(t)}}{\partial x_i} \right\|$, and define $\delta_t(i) \in \{0, 1\}$ as an indicator of whether x_i is unmasked at step t . The long-term effective gradient contribution of dimension i is then defined as:

$$\phi_i := \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T \delta_t(i) \cdot g_i^{(t)}.$$

This quantity reflects the accumulated importance of input dimension i over training. Dimensions that are frequently masked or provide inconsistent gradients across perturbations will exhibit suppressed values of ϕ_i , while those that maintain predictive gradients regardless of masking configurations will accumulate higher values. In effect, the model implicitly filters out unstable or co-dependent dimensions and shifts its representational capacity toward robust subspaces that sustain meaningful signal even under partial occlusion.

We interpret this process as the emergence of a perturbation-induced implicit attention mechanism. While no explicit attention weights are parameterized, we define a normalized proxy attention score $\alpha_i := \phi_i / \sum_{j=1}^d \phi_j$, which describes the model's long-term focus distribution over input dimensions. We refer to this behavior as *Perturbation-Induced Implicit Focus (PIIF)*. PIIF arises not from architectural design, but

from the interaction between input perturbations and gradient-based optimization, and serves as an effective filter that promotes stable, generalizable feature usage.

The training objective under this framework becomes an expectation over perturbed inputs:

$$\min_{\theta} \mathbb{E}_{(x,y) \sim \mathcal{D}} \mathbb{E}_{\tilde{x} = \mathcal{M}_p(x)} \mathcal{L}(f_\theta(\tilde{x}), y).$$

This objective enforces invariance under structured information loss, thereby guiding the model to establish predictive mappings that are insensitive to missing or irrelevant inputs. The perturbation operator $\mathcal{M}_p$ is easily implemented, incurs negligible overhead, and requires no changes to the model architecture or supervision strategy.

While conceptually related to techniques such as dropout and input masking, our method differs both in intention and effect. Dropout [4] seeks to prevent co-adaptation by injecting randomness into hidden activations, whereas our method introduces non-informative signals to drive attention toward invariant associations. Unlike masked modeling frameworks that reconstruct masked values, our method does not aim to predict missing content, but rather to train the model to ignore irrelevant parts. Moreover, whereas attention modules explicitly learn where to focus, PIIF emerges as a byproduct of training dynamics in perturbed environments. This property gives rise to representations that are not only compact and interpretable but also demonstrably more robust across domains and under partial input corruption.

IV. THEORETICAL ANALYSIS: PERTURBATION-INDUCED IMPLICIT FOCUS AND ROBUST FEATURE ASSOCIATION

This section provides a formal analysis of the mechanism by which structured random input perturbations induce an implicit feature selection and focus process in neural PDE solvers. Instead of relying on explicit attention modules, our method introduces semantic-free structured occlusions during training, compelling the model to identify and rely on stable input subspaces that remain informative across corrupted views. We show how this mechanism emerges naturally via gradient statistics and leads to improved generalization and robustness.

A. Gradient-Based Contribution and Implicit Feature Preference

To analyze how the network responds to perturbation, we track the gradient contribution of each input dimension. Let

$$g_i^{(t)} := \left\| \frac{\partial \mathcal{L}^{(t)}}{\partial x_i} \right\|$$

denote the gradient magnitude for input dimension x_i at training step t . Define the binary indicator:

$$\delta_t(i) := \begin{cases} 1, & x_i \text{ is not masked} \\ 0, & x_i \text{ is masked} \end{cases}.$$

We define the long-term average effective gradient contribution of x_i as:

$$\phi_i := \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T \delta_t(i) \cdot g_i^{(t)}.$$

This quantity reflects the network’s implicit preference for dimension i . Since $\mathbb{E}[\delta_t(i)] = 1 - p$, we obtain:

$$\mathbb{E}[\phi_i] = (1 - p) \cdot \mathbb{E} \left[g_i^{(t)} \mid \delta_t(i) = 1 \right].$$

B. From Perturbation Robustness to Focus Emergence

This formulation implies that input dimensions that are occasionally useful but fail to consistently contribute under random occlusions will see their effective gradient contribution ϕ_i decay over time. In contrast, those dimensions that maintain strong and stable gradient signals *across a variety of masking configurations* will accumulate higher values of ϕ_i , leading to stronger influence on model updates.

The model is thus implicitly encouraged to identify and prioritize those dimensions that form *robust associations* with the target output, even when other parts of the input are corrupted. This mechanism drives the emergence of perturbation-induced implicit focus (PIIF), whereby the network implicitly allocates its learning capacity to structurally robust, semantically relevant subspaces.

C. Robust Feature Association as Gradient Filtering

The perturbation process can also be viewed as a stochastic gate on input gradients. During training, the model repeatedly faces incomplete or partially corrupted inputs and must re-identify informative regions dynamically. This results in: Progressive suppression of unstable input dimensions and reinforcement of features that exhibit consistent predictive utility, leading to the gradual emergence of a sparse, task-relevant subspace guided by long-term gradient alignment.

This yields a filtering effect in the gradient space: the network shifts its attention—without any explicit attention layers—toward the most reliable dimensions.

D. Implicit Attention via Gradient Statistics

We interpret ϕ_i as a proxy for the attention weight of dimension i , and normalize it to define:

$$\alpha_i := \frac{\phi_i}{\sum_{j=1}^d \phi_j}, \quad \alpha_i \in [0, 1].$$

This distribution α reflects the implicit attention field learned by the model under stochastic perturbations. It encodes which dimensions the model chooses to rely on persistently, and which are ignored as unreliable noise.

E. Implications for Generalization and Robustness

The proposed mechanism directly supports the generalization and robustness of neural PDE solvers:

- **Generalization:** By focusing on features that maintain signal under diverse masking patterns, the model avoids overfitting to spurious correlations in the training data.
- **Robustness:** Exposure to corrupted inputs during training enables the model to better handle missing values, measurement noise, or boundary uncertainty during inference.

Training with structured random perturbations induces a form of implicit attention in neural networks. The model

learns to focus on features that maintain predictive relevance even when other inputs are partially corrupted. This behavior arises through a gradient-based filtering mechanism, whereby the network reinforces stable input-output associations and suppresses unstable ones. We refer to this mechanism as PIIF, which provides a strong theoretical foundation for the improved robustness and generalization observed in neural PDE solvers trained under this regime.

V. EXPERIMENTS

To evaluate the effectiveness of our proposed Perturbation-Induced Implicit Focus (PIIF) strategy, we conduct a series of experiments on representative partial differential equation (PDE) problems. Our evaluation focuses on two core aspects: (1) performance under limited training data to assess generalization and sample efficiency, and (2) robustness under partial input corruption to test the model’s resilience to incomplete or degraded inputs.

We use the Fourier Neural Operator (FNO) as the backbone architecture for all experiments, and compare our method against standard training (no perturbation), Dropout regularization, and Gaussian noise injection. Below, we present and analyze the key findings from two major experimental settings.

A. Generalization Under Varying Data Regimes

In this experiment, we study the ability of different methods to generalize when trained with limited data. We select two widely studied one-dimensional PDEs—the nonlinear Burgers equation and the linear advection equation—as testbeds. The number of training samples is varied across three settings: 32, 64, and 128. We evaluate model performance using several quantitative metrics, including root mean square error (RMSE), normalized RMSE (N RMSE), RMSE of conserved variables (RMSE C), RMSE at boundaries (RMSE B), and the maximum RMSE across the spatial domain.

The results (as shown in Table I) consistently demonstrate that our PIIF method outperforms standard training across nearly all settings. Notably, the performance advantage is especially pronounced in the low-data regimes. For instance, with only 32 training points, our method achieves a lower RMSE and significantly better RMSE C, indicating better preservation of the physical quantities governed by conservation laws. Similarly, in the 64-point setting, PIIF not only reduces the maximum prediction error but also improves boundary prediction accuracy—an aspect that is critical in solving PDEs with strong boundary effects.

As the training data size increases to 128 points, the gap between our method and standard training narrows, but PIIF still maintains a consistent edge, particularly in maximum error and boundary accuracy. This result suggests that the benefits of PIIF are most prominent when training data is scarce—a common scenario in scientific applications where data collection is expensive or simulations are costly. In short, PIIF serves as a powerful data-efficient inductive bias that guides the model toward learning more reliable, generalizable structures in the solution space.

TABLE I

PERFORMANCE COMPARISON OF STANDARD TRAINING AND OUR PIIF ON DIFFERENT NUMBERS OF TRAINING POINTS FOR THE 1D BURGERS AND 1D ADVECTION EQUATIONS. RMSE = ROOT MEAN SQUARE ERROR, N RMSE = NORMALIZED RMSE, RMSE C = RMSE OF CONSERVED VARIABLES, RMSE B = RMSE AT BOUNDARIES, MAX ERROR = MAXIMUM VALUE OF RMS ERROR.

Num	Training	1D Burgers equation					1D Advection equation				
		RMSE	N RMSE	RMSE C	RMSE B	Max Error	RMSE	N RMSE	RMSE C	RMSE B	Max Error
32	Standard	0.00506	0.00733	0.00884	0.00928	0.05485	0.12211	0.15733	0.01978	0.14811	0.76582
	Ours	0.00497	0.00702	0.00648	0.00905	0.04876	0.12293	0.15762	0.01997	0.13752	0.75491
64	Standard	0.00439	0.01057	0.00592	0.00836	0.01740	0.08974	0.11257	0.01456	0.11436	0.63521
	Ours	0.00416	0.00990	0.00475	0.00781	0.01467	0.08295	0.10924	0.01239	0.09841	0.59874
128	Standard	0.00204	0.00915	0.00045	0.01179	0.01523	0.06132	0.08745	0.00935	0.08578	0.51345
	Ours	0.00202	0.00926	0.00048	0.01094	0.01475	0.05928	0.08728	0.00922	0.08552	0.49646

TABLE II

ROBUSTNESS TO PARTIAL INPUT MASKING ON 2D POISSON EQUATION (RELATIVE ERROR %)

Method	0% Masking	10% Masking	20% Masking	30% Masking
Baseline (No Perturbation)	0.0172	0.0485	0.0947	0.1537
Dropout	0.0185	0.0392	0.0865	0.1421
Gaussian Noise Injection	0.0194	0.0396	0.0737	0.1349
Ours	0.0173	0.0321	0.0589	0.0977

B. Robustness to Partial Input Masking

In practical scientific modeling scenarios, input data is often incomplete, noisy, or partially corrupted. To simulate such settings, we perform robustness evaluation on the 2D Poisson equation at test time. Specifically, we simulate four levels of masking: 0%, 10%, 20%, and 30%, and measure the relative prediction error in each case.

As shown in Table II, our method consistently outperforms all baselines across varying levels of boundary condition masking. When no masking is applied, all methods achieve comparable performance. However, as more boundary information is removed, the error of the standard model increases significantly. While Dropout and Gaussian noise injection offer slight robustness improvements, they still exhibit a clear performance decline under heavier corruption. In contrast, our PIIF-trained model maintains much lower error across all masking levels and degrades more gracefully, demonstrating strong resilience to partial input loss. This performance trend highlights the unique advantage of PIIF: by introducing structured, semantically irrelevant perturbations during training, the model learns to ignore fragile, unstable input dimensions and instead focus on consistent and physically meaningful features. The perturbation-induced training dynamics encourage the development of representations that are robust to missing or occluded information, even without explicit architectural constraints or supervision on masked inputs.

These findings strongly support our hypothesis that structured irrelevance acts as an effective inductive prior, guiding neural PDE solvers to learn not only more generalizable, but also more resilient solution representations. The above experiments validate the central claims of our work from both generalization and robustness perspectives: Under low-data regimes, PIIF consistently improves prediction accuracy

across multiple PDEs, demonstrating strong sample efficiency and effective inductive regularization. Under partial input corruption, PIIF significantly outperforms standard and dropout-based baselines, showing improved robustness and graceful performance degradation as information loss increases. These benefits are achieved without modifying the model architecture or requiring additional supervision, attesting to the generality and practicality of the PIIF strategy.

Together, these results suggest that training neural PDE solvers with structured perturbation is a simple yet powerful way to promote both accurate and robust learning in challenging scientific machine learning settings.

VI. FUTURE WORK

With the continued advancement of deep learning [26], especially in its application to scientific computing and partial differential equation (PDE) modeling, new opportunities have emerged for developing principled and data-efficient learning frameworks. In this context, the proposed PIIF framework has demonstrated strong generalization and robustness across a range of PDE-solving tasks. Nevertheless, several promising directions remain for future research. First, we plan to explore more advanced data augmentation techniques [12], [15]–[19], [21] to enhance model robustness under diverse conditions. These methods aim to further regularize neural PDE solvers in the presence of sparse supervision, noisy inputs, or structural biases. In particular, we intend to tailor such augmentations to the spatial and physical characteristics of scientific problems, including domain warping, boundary reshaping, and function-level interpolation. Second, we plan to integrate adversarial learning strategies [13], [14], [17], [20], [22], [23] into the PIIF framework. Generating input perturbations in an adversarial manner can drive the model to identify and defend against

subtle but harmful correlations, thereby reinforcing its focus on physically consistent and generalizable features. Recent advances in cross-modality perturbations and attention-shift-based adversarial training provide exciting opportunities to extend this line of work within scientific machine learning.

In addition, we are interested in extending PIIF to more complex or uncertain problem settings, such as inverse problems, multi-physics coupling, and uncertainty quantification. These scenarios often involve incomplete observations, distributional shifts, or noisy boundary conditions, making them well-suited for the inductive bias introduced by our perturbation-driven learning scheme. We believe that combining PIIF with structured augmentation, adversarial robustness, and domain-specific insights will further enhance the robustness, interpretability, and applicability of neural PDE solvers in real-world scientific and engineering applications.

VII. CONCLUSION

We have introduced a simple yet powerful training strategy for neural PDE solvers based on injecting structured, semantically irrelevant perturbations into the input space. Unlike conventional data augmentation or regularization techniques, our method induces an implicit feature selection mechanism—termed Perturbation-Induced Implicit Focus (PIIF)—that guides the network to attend to stable, physically meaningful features across corrupted inputs. Experiments demonstrate that our approach improves generalization to unseen domains, robustness to partial input corruption, and alignment with physical laws, all without requiring architectural changes or auxiliary supervision. These findings suggest that structured irrelevance offers a novel and effective inductive bias for scientific machine learning.

Looking ahead, we believe this perturbation-driven focus mechanism can be further extended to other domains involving spatial-temporal dynamics, uncertainty quantification, and physics-informed learning, opening new directions for robust and interpretable neural solvers.

REFERENCES

- [1] M. Raissi, P. Perdikaris, and G. E. Karniadakis, “Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations,” *Journal of Computational Physics*, vol. 378, pp. 686–707, 2019.
- [2] Z. Li, N. Kovachki, K. Azizzadenesheli, B. Liu, A. Bhattacharya, A. Stuart, and A. Anandkumar, “Fourier neural operator for parametric partial differential equations,” *International Conference on Learning Representations (ICLR)*, 2021.
- [3] J. Brandstetter, G. A. Holzmann, M. W. Dusenberry, M. Dehghani, and B. S. Schoelkopf, “Message passing neural PDE solvers,” *International Conference on Learning Representations (ICLR)*, 2022.
- [4] N. Srivastava, G. Hinton, A. Krizhevsky, I. Sutskever, and R. Salakhutdinov, “Dropout: A simple way to prevent neural networks from overfitting,” *Journal of Machine Learning Research*, vol. 15, no. 1, pp. 1929–1958, 2014.
- [5] J. Devlin, M. Chang, K. Lee, and K. Toutanova, “BERT: Pre-training of deep bidirectional transformers for language understanding,” *Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, 2019.
- [6] A. Vaswani, N. Shazeer, N. Parmar, et al., “Attention is all you need,” *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.
- [7] K. Clark, U. Khandelwal, O. Levy, and C. D. Manning, “What does BERT look at? An analysis of BERT’s attention,” *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2019.
- [8] A. Shrikumar, P. Greenside, A. Shcherbina, and A. Kundaje, “Not just a black box: Learning important features through propagating activation differences,” *International Conference on Machine Learning (ICML)*, 2017.
- [9] J. Zhu, S. Zabarar, P. Perdikaris, “Physics-constrained deep learning for high-dimensional surrogate modeling and uncertainty quantification without labeled data,” *Journal of Computational Physics*, vol. 394, pp. 56–81, 2019.
- [10] M. Tancik, P. Srinivasan, B. Mildenhall, et al., “Fourier features let networks learn high frequency functions in low dimensional domains,” *NeurIPS*, 2020.
- [11] M. M. Bronstein, J. Bruna, T. Cohen, and P. Veličković, “Geometric deep learning: Grids, groups, graphs, geodesics, and gauges,” *arXiv preprint arXiv:2104.13478*, 2021.
- [12] Y. Gong, L. Huang, and L. Chen, “Eliminate deviation with deviation for data augmentation and a general multi-modal data learning method,” *arXiv preprint arXiv:2101.08533*, 2021.
- [13] Y. Gong, L. Huang, and L. Chen, “Person re-identification method based on color attack and joint defence,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, pp. 4313–4322, 2022.
- [14] Y. Gong, Z. Zhong, Y. Qu, Z. Luo, R. Ji, and M. Jiang, “Cross-Modality Perturbation Synergy Attack for Person Re-identification,” in *Proc. NeurIPS*, 2024.
- [15] Y. Gong, J. Li, L. Chen, and M. Jiang, “Exploring color invariance through image-level ensemble learning,” *arXiv preprint arXiv:2401.10512*, 2024.
- [16] J. Lin, Z. Wang, Y. Hou, Y. Tang, and M. Jiang, “Phy124: Fast physics-driven 4d content generation from a single image,” *arXiv preprint arXiv:2409.07179*, 2024.
- [17] Y. Gong, Y. Hou, Z. Wang, Z. Lin, and M. Jiang, “Adversarial Learning for Neural PDE Solvers with Sparse Data,” *arXiv preprint arXiv:2409.02431*, 2024.
- [18] Y. Gong, C. Zhang, Y. Hou, L. Chen, and M. Jiang, “Beyond dropout: Robust convolutional neural networks based on local feature masking,” in *Proc. Int. Joint Conf. Neural Netw. (IJCNN)*, pp. 1–8, 2024.
- [19] Y. Gong, Y. Hou, C. Zhang, and M. Jiang, “Beyond augmentation: Empowering model robustness under extreme capture environments,” in *Proc. Int. Joint Conf. Neural Netw. (IJCNN)*, pp. 1–8, 2024.
- [20] Y. Gong, Q. Zeng, D. Xu, Z. Wang, and M. Jiang, “Cross-modality attack boosted by gradient-evolutionary multifunction optimization,” *arXiv preprint arXiv:2409.17977*, 2024.
- [21] Y. Gong, Z. Zeng, and F. Ye, “Person re-identification method based on grayscale feature enhancement,” *J. Comput. Appl.*, vol. 41, no. 12, p. 3590, 2021.
- [22] Q. Zeng, Y. Gong, and M. Jiang, “Cross-Task Attack: A Self-Supervision Generative Framework Based on Attention Shift,” in *Proc. Int. Joint Conf. Neural Netw. (IJCNN)*, pp. 1–8, 2024.
- [23] J. Lin, Z. Wang, D. Xu, S. Jiang, Y. Gong, and M. Jiang, “Phys4DGen: Physics-Compliant 4D Generation with Multi-Material Composition Perception,” *arXiv preprint arXiv:2411.16800*, 2024.
- [24] H. Wu, Y. Yang, A. I. Aviles-Rivero, J. Ren, S. Chen, H. Chen, and L. Zhu, “Semi-supervised video desnowing network via temporal decoupling experts and distribution-driven contrastive regularization,” in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, pp. 70–89, 2024.
- [25] L. Peng, Y. Wang, X. Di, X. Fu, Y. Cao, Z.-J. Zha et al., “Boosting image de-raining via central-surrounding synergistic convolution,” in *Proc. AAAI Conf. Artif. Intell.*, vol. 39, no. 6, pp. 6470–6478, 2025.
- [26] Z. Yang, D. Wu, W. Zhang, B. Li, and W. Wang, “Handling label uncertainty for camera incremental person re-identification,” in *Proc. 31st ACM Int. Conf. Multimedia (MM)*, pp. 6253–6263, 2023.